

Project Name: **YVAM-USA**

Unit Tag: **CH-4**

Qty.: **1**

Model: **YVAM1650**

Full Load - Design

PIN

YVAM1650A2	646BZHC7XB	SAXLXUSNX1	XXXXXRXM4M	2W1SXXA2B4	TDH1016XXX	XSNSGXVXX	XXXXXXEAS2	914D481XXX
...5...10	...5...20	...5...30	...5...40	...5...50	...5...60	...5...70	...5...80	...5...90
XXXXXXXXIXN	XMPG257290	1180457260	XXX60A0XX					
...5...100	...5...110	...5...120	...5...130					

Unit

Model No.	YVAM1650
Number of Compressors	1
Compressor Type	Variable Speed
Number of Compressor Circuits	1
Capacity Control	10% - 100%
Refrigerant Type	R-1234ze
Compressor	M2C-233FAC
Cooling Mode Operation	
Cooling Capacity [kW]	1607
Total Input Power [kW]	427.5
Cooling [kW/kW]	3.759
NPLV.IP [kW/kW]	12.58
A-Weighted Sound Power [dB(A)]	102.5
Electrical Data	
Nominal Voltage / Voltage Limits	460/3/60 / 414V - 508V
Min. Circuit Ampacity [A]	708.0
Max. Fuse / CB Rating [A]	1000.0
Unit Short Circuit Withstand (STD) [kA]	10
Wires Per Phase	4
Wire Range (Lug Size)	4/0 - 500 kcmil
RLA [A]	590.6
Control kVA	5
Compressor kW	362.4



Performance Impacting Options

Fan Type	High airflow low sound EC fans with vaned diffuser and harmonic filter
----------	--

Weight & Dimensional Data

Shipping Weight [kg]	10626
Operating Weight [kg]	10998
Refrigerant Charge [kg]	340
Overall Length [mm]	15244
Overall Width [mm]	2235
Overall Height [mm]	2503

Project Name: YVAM-USA

Unit Guid: 1cf145e4-5cc7-4e7f-9a56-88872310980d

Rating Engine Version: 1.0.9662

Unit Name: CH-4

Version: SN26.06

CHL.2026-06.002

Generated: 2026/07/06 at 00:28

Page 1 of 5



Performance Report

Performance Specification

Page 2 of 5

Project Name: YVAM-USA

Unit Tag: CH-4

Qty.: 1

Model: YVAM1650

Heat Exchanger Performance

Evaporator		Condenser (Air Cooled)	
Model	EA2914-481-DS-2	Ambient Air Temperature* [°C]	48.0
Entering Fluid Temperature* [°C]	32.00	Altitude* [m]	0.00
Leaving Fluid Temperature* [°C]	22.00	Number of Fans	26
Flow Rate [L/s]	40.16	Total Air Flow [m3/s]	207
Fouling Factor* [m2 K/kW]	0.0176	Total Fan Power [kW]	65.16
Fluid Type* / Concentration* [%]	Propylene Glycol / 25.0	External Static Pressure [Pa]	0
Passes*	2		
Pressure Drop [kPa]	35.0		
Minimum Flow Rate [L/s]	29.95		
Maximum Flow Rate [L/s]	81.83		

* Designates user specified input

Outside the scope of AHRI Air-Cooled Water-Chilling Packages Certification Program or not optionally certified, but is rated in accordance with AHRI Standard 550/590(I-P) and AHRI Standard 551/591(SI). Auxiliary components included in total kW - Chiller controls.

Part Load Performance (Based on Air Cooled Efficiency Map w/VPF)

Ambient [°C]	Percent Load					
	100.0	90.0	80.0	70.0	60.0	50.0
48.0	3.759	3.867	3.894	3.867	3.823	3.699
46.0	4.062	4.165	4.199	4.186	4.138	3.995
44.0	4.375	4.488	4.533	4.515	4.466	4.333
42.0	4.699	4.832	4.886	4.879	4.882	4.815
40.0	5.058	5.209	5.293	5.326	5.347	5.262
38.0	5.452	5.632	5.737	5.805	5.833	5.781

Heat map shows efficiency values ranging from 60% higher than design condition efficiency (blue) to best efficiency (green).

Project Name: YVAM-USA

Unit Guid: 1cf145e4-5cc7-4e7f-9a56-88872310980d

Rating Engine Version: 1.0.9662

Unit Name: CH-4

Version: SN26.06

CHL.2026-06.002

Generated: 2026/07/06 at 00:28

Page 2 of 5



Performance Report

Performance Specification

Page 3 of 5

Project Name: YVAM-USA

Unit Tag: CH-4

Qty.: 1

Model: YVAM1650

Part Load Performance (Based on Air Cooled Efficiency Map w/VPF)

Percent Load	Ambient [°C]	Capacity [kW]	Power Input [kW]	Unit Efficiency [kW/kW]	EFT [°C]	LFT [°C]	Flow Rate [L/s]	Evap. PD [kPa]
100.0	48.0	1607	427.5	3.759	32.00	22.00	40.16	35.0
100.0	46.0	1607	395.7	4.062	32.00	22.00	40.16	35.0
100.0	44.0	1607	367.4	4.375	32.00	22.00	40.16	35.0
100.0	42.0	1607	342.0	4.699	32.00	22.00	40.16	35.0
100.0	40.0	1607	317.7	5.058	32.00	22.00	40.16	35.0
100.0	38.0	1607	294.8	5.452	32.00	22.00	40.16	35.0
90.0	48.0	1447	374.0	3.867	32.00	22.00	36.15	28.6
90.0	46.0	1447	347.3	4.165	32.00	22.00	36.15	28.6
90.0	44.0	1447	322.3	4.488	32.00	22.00	36.15	28.6
90.0	42.0	1447	299.4	4.832	32.00	22.00	36.15	28.6
90.0	40.0	1447	277.7	5.209	32.00	22.00	36.15	28.6
90.0	38.0	1447	256.9	5.632	32.00	22.00	36.15	28.6
80.0	48.0	1286	330.2	3.894	32.00	22.00	32.13	22.7
80.0	46.0	1286	306.2	4.199	32.00	22.00	32.13	22.7
80.0	44.0	1286	283.7	4.533	32.00	22.00	32.13	22.7
80.0	42.0	1286	263.2	4.886	32.00	22.00	32.13	22.7
80.0	40.0	1286	242.9	5.293	32.00	22.00	32.13	22.7
80.0	38.0	1286	224.1	5.737	32.00	22.00	32.13	22.7
70.0	48.0	1125	291.0	3.867	31.39	22.00	29.95	19.7
70.0	46.0	1125	268.8	4.186	31.39	22.00	29.95	19.7
70.0	44.0	1125	249.2	4.515	31.39	22.00	29.95	19.7
70.0	42.0	1125	230.6	4.879	31.39	22.00	29.95	19.7
70.0	40.0	1125	211.2	5.326	31.39	22.00	29.95	19.7
70.0	38.0	1125	193.8	5.805	31.39	22.00	29.95	19.7
60.0	48.0	964.4	252.3	3.823	30.05	22.00	29.95	19.7
60.0	46.0	964.4	233.0	4.138	30.05	22.00	29.95	19.7
60.0	44.0	964.4	215.9	4.466	30.05	22.00	29.95	19.7
60.0	42.0	964.4	197.5	4.882	30.05	22.00	29.95	19.7
60.0	40.0	964.4	180.3	5.347	30.05	22.00	29.95	19.7
60.0	38.0	964.4	165.3	5.833	30.05	22.00	29.95	19.7
50.0	48.0	803.6	217.2	3.699	28.71	22.00	29.95	19.7
50.0	46.0	803.6	201.2	3.995	28.71	22.00	29.95	19.7
50.0	44.0	803.6	185.5	4.333	28.71	22.00	29.95	19.7
50.0	42.0	803.6	166.9	4.815	28.71	22.00	29.95	19.7
50.0	40.0	803.6	152.7	5.262	28.71	22.00	29.95	19.7
50.0	38.0	803.6	139.0	5.781	28.71	22.00	29.95	19.7

Project Name: YVAM-USA

Unit Guid: 1cf145e4-5cc7-4e7f-9a56-88872310980d

Rating Engine Version: 1.0.9662

Unit Name: CH-4

Version: SN26.06

CHL2026-06.002

Generated: 2026/07/06 at 00:28

Page 3 of 5



Performance Report

Performance Specification

Project Name: YVAM-USA

Unit Tag: CH-4

Qty.: 1

Model: YVAM1650

Part Load Performance Condenser

Percent Load	Ambient [°C]	Heat Rejection [kW]	Discharge Air Temperature [°C]	Total Air Flow [m3/s]	Active Fans
100.0	48.0	1998	56.17	207	26
100.0	46.0	1978	54.78	192	26
100.0	44.0	1953	53.06	184	26
100.0	42.0	1928	51.06	181	26
100.0	40.0	1908	49.33	174	26
100.0	38.0	1885	47.33	172	26
90.0	48.0	1795	56.28	183	26
90.0	46.0	1773	54.61	175	26
90.0	44.0	1748	52.61	173	26
90.0	42.0	1726	50.61	170	26
90.0	40.0	1704	48.61	168	26
90.0	38.0	1686	46.89	161	26
80.0	48.0	1590	55.72	174	26
80.0	46.0	1568	53.89	169	26
80.0	44.0	1549	52.17	162	26
80.0	42.0	1528	50.06	162	26
80.0	40.0	1510	48.17	158	26
80.0	38.0	1490	46.17	156	26
70.0	48.0	1390	55.11	165	26
70.0	46.0	1369	53.17	163	26
70.0	44.0	1354	51.50	154	26
70.0	42.0	1332	49.17	158	26
70.0	40.0	1315	47.28	154	26
70.0	38.0	1301	45.61	145	26
60.0	48.0	1192	54.50	156	26
60.0	46.0	1177	52.83	147	26
60.0	44.0	1161	50.83	145	26
60.0	42.0	1142	48.72	145	26
60.0	40.0	1128	47.06	136	26
60.0	38.0	1113	44.94	137	26
50.0	48.0	1002	54.06	140	26
50.0	46.0	986.4	52.06	138	26
50.0	44.0	971.3	50.06	136	26
50.0	42.0	954.3	48.17	131	26
50.0	40.0	941.4	46.28	128	26
50.0	38.0	927.4	44.17	128	26

Project Name: YVAM-USA

Unit Guid: 1cf145e4-5cc7-4e7f-9a56-88872310980d

Rating Engine Version: 1.0.9662

Unit Name: CH-4

Version: SN26.06

CHL2026-06.002

Generated: 2026/07/06 at 00:28



Performance Report

Performance Specification

Page 5 of 5

Project Name: **YVAM-USA**

Unit Tag: **CH-4**

Qty.: **1**

Model: **YVAM1650**

Notes:

Country of Origin: Mexico

Displacement Power Factor refers to compressor only. Unit Power Factor depends on fan option selected. Calculated value is available by request.

Minimum and maximum evaporator flow information are for full load ratings with Water.

The product image shown is for illustrative purposes only and is not representative of selected options.

Standard Single Point, Dual Aux Box with UPS, with Dual CB's option is 10kA unless upstream circuit breaker is rated for 100kA interrupting current